

나현준 (Hyun June Na)

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EDUCATION

Seoul National University, B.S. in Mechanical Engineering

2022.03 – Present

PROJECTS

WAM (Weightlifting Assistant Machine) — Clamp-on Strength Training Assist Device 2026.03 – 2026.06

Mechatronics Course Team Project (Team 10, 5 members) — Hardware Lead

- Led mechanical design and fabrication of a clamp-on gym-machine assist device, owning the full hardware path from concept to working prototype
- Redesigned the drive mechanism from a rack-and-pinion actuator to a drum-and-strap cable-winding system to cut weight and enable attach/detach on existing machines
- Modeled the motor-drum-strap power transmission in CAD and fabricated the full enclosure and mechanical assembly
- Devised an adjustable clamp mount compatible with nearly all standard "plate & pin" gym machines, the key differentiator from single-machine built-in competitors
- Sized the system to a low-cost motor under budget, achieving stable 6 kg assistance on a self-built test rig with headroom to scale via a stronger motor